

Name: _____

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24 points possible; each part is worth 2 points. No aids (book, notes, calculator, phone, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form.

1. [12 points] Compute the derivatives of the following functions.

a. $f(x) = x^2 e^{x^2+1} + x + 1$

b. $f(x) = \tan(2x) + \sin^2(x) \cos(x)$

c. $f(x) = \frac{3x^2 - 1}{\ln(x^2 + 1)}$

d. $f(x) = \sqrt{x + \sin^2(x^2)}$

e. $f(x) = e^{1/x} - \frac{x^2}{2} + \frac{2}{\sqrt{x}}$

f. $f(x) = \ln(a + b^{2x})$ where a and b are constants

2. [12 points] Compute the following antiderivatives (indefinite integrals) and definite integrals. Remember that antiderivatives need a “+C”.

a. $\int_{-2}^2 \sqrt{4-x^2} \, dx$

b. $\int_0^3 (x+2)(x-3) \, dx$

c. $\int 2x^3 \sqrt{x^2+1} \, dx$

d. $\int_0^{\pi/3} \sec^2(x) \tan(x) \, dx$

e. $\int \frac{e^x}{1 + e^{(2x)}} \, dx$

f. $\int \tan(2x) \, dx$